4

Shohel Ahmed

Professional Experience:

- Islamic University of Technology (IUT)
  Assistant Professor (Feb'2020 ~)
- Military Institute of Science & Technology
  Assistant Professor (Sep'19 – Jan'20)
- Kona Software Lab Ltd.
  Sr. Project Manager (Jul'16 ~ Sep'19)
- Samsung R&D Institute Bangladesh (SRBD)
  Chief Engineer & Group Leader (Jan'15~ Jul'16)
  Digital Imaging
  Technical Lead (Dec'10~ Dec'14)
  Platform Lab
- Sylhet International University
  Assistant Professor (Jul'09 – Nov'10)
  Lecturer (Apr'04 – Jul'04)
- Inha University, Korea
  Research Assistant (Aug'07-Jun'09)
  Intelligent Mobile Lab

Educational Experience:

- Master of Engineering in Information Engineering
  Inha University, Incheon, South Korea
- BSc in Computer Science & Information Technology
  Islamic University of Technology, Gazipur, Bangladesh
- Higher Secondary Certificate (HSC)
  Secondary School Certificate (SSC)
  Rangpur Cadet College, Rangpur, Bangladesh

Patents:

- ✓ Shohel ahmed, Shake Md. Riazul Islam, Kwak Kyung Sup - " METHOD AND SYSTEM FOR FINGERPRINT-BASED PAYMENT " Application No.(Date) 1020170141375 (2017.10.27),Publication : KR101951125B1
- ✓ BARI, A.S.M. Hossain, Shohel AHMED ,AHSAN, Shegufta Bakht, Sadre Ala PARVEZ,AZIZ, Syeda Persia, Md. Tawhidul Islam CHOWDHURY- "ELECTRONIC DEVICE AND METHOD FOR PROVIDING INFORMATION THEREOF" Publication number: 20150220247

For More info https://cse.iutoic-dhaka.edu/profile/sohel/

5

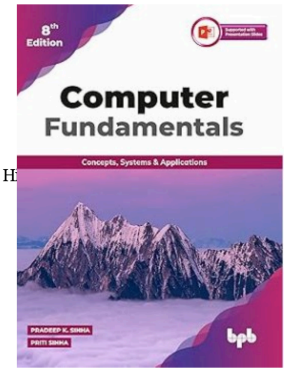
Important For you

Course Code: SWE 4101

Course Name: Introduction to Software Engineering

Recommended Books:

- P. K. Sinha and P. Sinha, *Computer Fundamentals*, 8th ed. BPB Publications, 2020
- R. S. Pressman and B. R. Maxim, *Software Engineering: A Practitioner's Approach*, 8th ed. McGraw-Hi



Following the lecture slides & class will be good enough

6

Assessment & Grading Policy

Course Assessment Method

10% participation	30% Quizzes and assignment	30% Midterm	30% final

Grading Policy

Marks out of 100	Letter Grade	Grade Points	Marks out of 100	Letter Grade	Grade Point
80 - 100	A+	4.00	55 - 59	B-	2.75
75 - 79	A	3.75	50 - 54	C+	2.50
70 - 74	A-	3.50	45 - 49	C	2.25
65 - 69	B+	3.25	40 – 44	D	2.00
60 - 64	B	3.00	00 - 39	F	0.00

10

Lecture Plan

Weeks	Topics
1	Introduction to Computers and Basic Concepts
2	Hardware Components and Peripherals
3	Computer Architecture & Number Systems
4	Software Concepts – Types and System Software
5	Programming and Language Translators
6	Networking and the Internet
7	Web and Emerging Technologies
8	Introduction to SW Engineering, Software Development Life Cycle (SDLC)
9	Software Process Models
10	Software Requirements Engineering, UI/UX
11	Basics of SW design
12	SW Testing
13	Software Documentation
14	Software Release and Maintenance

11

Meet Your Digital Brain: The Computer


Desktop


Laptop


Mobile


ATM


Point of Sale (POS)


Samsung Gear 360



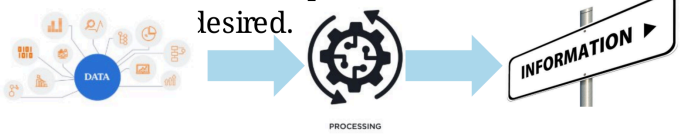
Samsung NX1

- √ A computer is an electronic device thatakes input, processes it, stores data, and produces outputto help us perform tasks.
- √ Your smartphone today is millions of times more powerful than computers from the 1960s!

12

Computer

- √ The word computer comes from the word “compute” which means, “to calculate”.
- √ Thereby, a computer is an electronic device that can perform arithmetic operations at high speed.
- √ A computer is also called adata processor beca[illegible]re, process, and retrieve d[illegible]lesired.

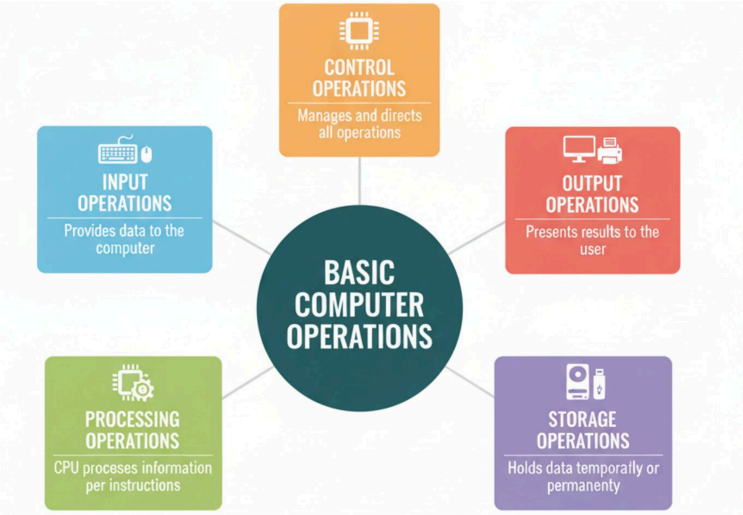


The activityof processing data using a computer is called data processing

What are the activities? Capture Data Manipulate Data Output Results

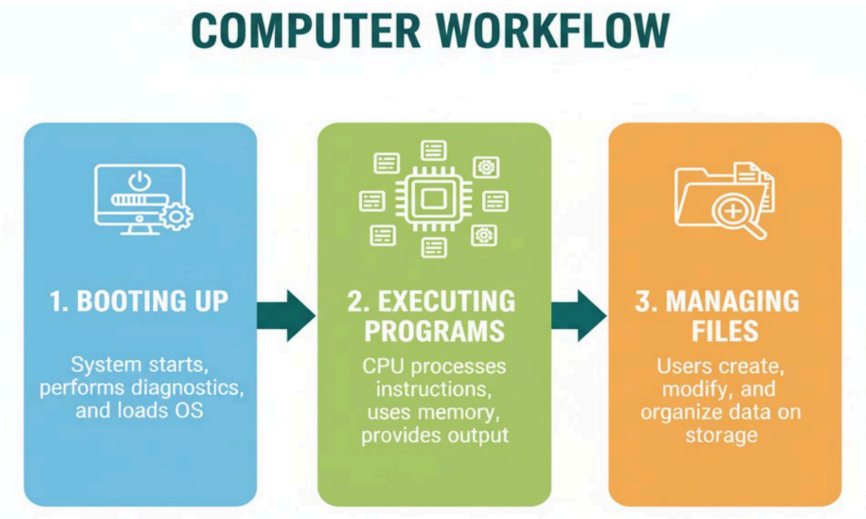
16

Basic Computer Operations



17

Computer Workflow



18

Events after you press the power button

√ Power-On Self Test (POST): The computer performs a self-check to ensure all hardware components are functioning correctly.

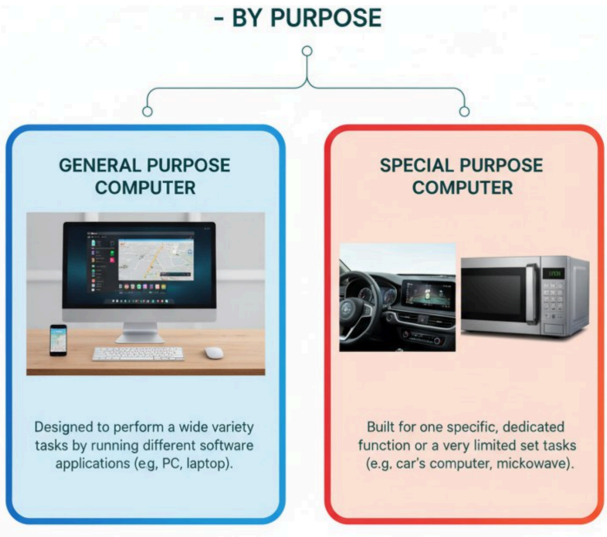
√ Boot Loader: The computer loads a small program called the bootloader, which in turn loads the operating system into memory.

√ Operating System: The operating system initializes various hardware components and prepares the system for use.

√ User Interface Finally, the user interface (such as Windows Desktop or MacOS Finder) is displayed, allowing you to interact with the computer.

22

Types of Computers (by Purpose)



23

Limitations of Computer

No IQ
Can not think creatively or understand without input.

Vulnerability to Cyber Threats
Exposed to cyber threats like hacking, viruses, and malware, risking data integrity, operations, and finances.

Lack of Decision Making
They can analyze data but cannot make subjective or moral judgments.

Ethical and Moral Reasoning
computers cannot reason ethically or morally; they can only follow human-defined guidelines.

Human Dependency
Need Human programming, instructions and data input.

No feelings
Can not understand or express human feelings/empathy.